

War Card Game Simulation

Code Documentation and Analysis

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Course: CST-180: Python Programming 1

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Date: September 24, 2025

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War Card Game Simulation Documentation

Part 1: Problem Statement

This program simulates a modified version of the classic War card game between two players over 10 complete matches. The simulation models a simplified card game where each player receives an identical deck of 26 cards containing values 1 through 13, with each value appearing exactly twice. This creates a fair and balanced starting condition for both players.

The game operates through a series of 10 independent matches. In each match, both players' decks are randomly shuffled to ensure unpredictability and prevent any strategic advantage from previous games. Each match consists of 26 rounds, where both players simultaneously reveal one card from their deck. The player with the higher-valued card wins that round, and their round win count increments by one.

The core mechanic of the game is straightforward card comparison. When two cards are revealed, the player with the numerically higher card value wins the round. If both cards have identical values, the round results in a tie, and neither player receives a point for that round. This tie mechanism adds an element of chance that reflects the random nature of card distribution.

After all 26 rounds of a match are completed, the player who won more rounds is declared the winner of that match. The match winner's total match win count increases by one. If both players won an equal number of rounds within a match, the entire match is recorded as a tie.

The simulation tracks comprehensive statistics throughout all 10 matches, including total match wins for each player, the cumulative number of ties (both round ties and match ties), and calculates win percentages for both players based on their match victories. The primary goal is to demonstrate basic programming concepts including loops, conditionals, list manipulation, randomization, and statistical calculation while creating a functional and engaging card game simulation.

Part 2: Summary of Logic (Input-Processing-Output)

Input: The simulation uses predefined game parameters including deck composition (values 1-13, duplicated for 26 total cards), number of matches (10), and rounds per match (26). The primary input mechanism is Python's random shuffling function, which randomizes the card order for each player at the start of every match. No user input is required during execution.

Processing: The simulation follows a hierarchical loop structure. The outer loop iterates through 10 matches, and within each match, an inner loop processes 26 rounds of card comparison. At the start of each match, both decks are shuffled independently. For each round, the program draws one card from each player's deck, compares their values, and updates the appropriate win counter. After completing all 26 rounds, the program determines the match winner by comparing round win totals. Throughout execution, the program maintains running totals of match wins and ties, updating these statistics after each match conclusion.

Output: The program generates real-time output displaying the progression of each game. During each round, it prints which player won and their current round win count, or announces a tie if applicable. After each match, it declares the match winner. Upon completing all 10 matches, the program displays comprehensive final statistics including total match wins for each player, total number of ties, and win percentages calculated as $(\text{match wins} / 10 \text{ matches}) \times 100$.

Part 3: Detailed Pseudocode

Variable Initialization

```
// PROGRAM War Card Game Simulation
// This program simulates 10 matches of a war card game between two players.
// Each player has an identical deck of 26 cards (values 1-13, each appearing twice).

// Card Deck Definition
DEFINE deck_of_cards as [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13]
DEFINE player_1_deck as deck_of_cards + deck_of_cards // 26 cards total
DEFINE player_2_deck as deck_of_cards + deck_of_cards // 26 cards total

// Game Tracking Variables
DEFINE player_1_round_wins as 0 // Round wins in current match
DEFINE player_2_round_wins as 0 // Round wins in current match
DEFINE player_1_match_wins as 0 // Total match wins across all games
DEFINE player_2_match_wins as 0 // Total match wins across all games
DEFINE total_ties as 0 // Cumulative ties (rounds + matches)

// Statistical Variables
DEFINE player_1_win_percentage as 0.0 // Player 1's final win percentage
DEFINE player_2_win_percentage as 0.0 // Player 2's final win percentage
```

Main Game Loop Structure

```
// Outer Loop: Play 10 Matches
FOR match FROM 1 TO 10

    // Match Initialization
    PRINT "=== Game", match, "==="

    // Step 1: Shuffle both decks for this match
    RANDOM SHUFFLE player_1_deck
    RANDOM SHUFFLE player_2_deck

    // Step 2: Reset round counters for new match
    SET player_1_round_wins to 0
    SET player_2_round_wins to 0
```

Part 3: Detailed Pseudocode (Continued)

Round Processing Loop

```
// Inner Loop: Play 26 Rounds per Match
FOR round_num FROM 1 TO 26

    // Step 3: Draw cards for this round
    // (Index is round_num - 1 because arrays start at 0)
    SET player_1_card to player_1_deck[round_num - 1]
    SET player_2_card to player_2_deck[round_num - 1]

    // Step 4: Compare cards and update round winners
    IF player_1_card > player_2_card THEN
        // Player 1 wins this round
        INCREMENT player_1_round_wins by 1
        PRINT "Player 1 wins round count:", player_1_round_wins

    ELSE IF player_2_card > player_1_card THEN
        // Player 2 wins this round
        INCREMENT player_2_round_wins by 1
        PRINT "Player 2 wins round count:", player_2_round_wins

    ELSE
        // Cards are equal - tie round
        INCREMENT total_ties by 1
        PRINT "Tie - no one wins this round"
    ENDIF

ENDFOR // End of 26 rounds
```

Part 3: Detailed Pseudocode (Continued)

Match Winner Determination

```
// Step 5: Determine match winner after 26 rounds
IF player_1_round_wins > player_2_round_wins THEN
    // Player 1 won more rounds in this match
    INCREMENT player_1_match_wins by 1
    PRINT "Player 1 wins game!"

ELSE IF player_2_round_wins > player_1_round_wins THEN
    // Player 2 won more rounds in this match
    INCREMENT player_2_match_wins by 1
    PRINT "Player 2 wins game!"

ELSE
    // Both players won equal rounds - match tie
    INCREMENT total_ties by 1
    PRINT "Game is a tie!"
ENDIF

ENDFOR // End of 10 matches
```


Final Statistics Calculation

```
// Step 6: Calculate final win percentages
COMPUTE player_1_win_percentage as (player_1_match_wins / 10) * 100
COMPUTE player_2_win_percentage as (player_2_match_wins / 10) * 100

// Step 7: Display final statistics
PRINT "Player 1 total wins:", player_1_match_wins
PRINT "Player 2 total wins:", player_2_match_wins
PRINT "Total ties:", total_ties
PRINT "Player 1 win percentage:", player_1_win_percentage, "%"
PRINT "Player 2 win percentage:", player_2_win_percentage, "%"

// END PROGRAM
```

Part 4: Test Data Processing and Results Validation

Sample Simulation Run Analysis

Due to the random nature of the card shuffling, each execution produces different results. However, the underlying mechanics remain consistent and predictable. Below is an analysis of a representative simulation run:

Match 1 Example Breakdown:

- Both decks shuffled randomly at match start
- 26 rounds played, each comparing one card from each deck
- Example progression: Player 1 might win rounds 1, 3, 7, etc.
- Final round count determines match winner
- Typical match outcome: 13-11, 14-10, or similar distributions

Expected Statistical Patterns:

Metric	Expected Range	Reasoning
Player 1 Match Wins	0-10	Random shuffling creates variable outcomes
Player 2 Match Wins	0-10	Random shuffling creates variable outcomes
Total Matches	10	Fixed by program design
Win Percentage	0-100%	Depends on match wins (wins/10 × 100)
Round Ties	15-35	Approximately 2-3 ties per match expected

Mathematical Verification:

The probability of a tie in any given round can be calculated:

- Each deck has 2 cards of each value (1-13)
- When both players draw the same value, a tie occurs
- Expected tie probability per round $\approx 2/26 \times 2/26 = 0.0059$ or $\sim 0.6\%$
- Over 260 rounds (10 matches × 26 rounds), expect ~ 1 -2 ties per match
- Actual implementation may show slightly different patterns due to shuffling mechanics

Part 4: Test Data Processing and Results Validation (Continued)

Win Percentage Calculation Verification

Test Case 1: Player 1 Wins 7 Matches

- Match wins: 7
- Calculation: $(7 / 10) \times 100 = 70.0\%$
- Expected output: "Player 1 win percentage: 70.0%"
- Verification: ✓ Correct

Test Case 2: Even Split (5-5)

- Player 1 match wins: 5
- Player 2 match wins: 5
- Player 1 percentage: $(5 / 10) \times 100 = 50.0\%$
- Player 2 percentage: $(5 / 10) \times 100 = 50.0\%$
- Expected output: Both players at 50.0%
- Verification: ✓ Correct

Test Case 3: Dominant Victory (9-1)

- Winner: 9 matches, Loser: 1 match
- Winner percentage: $(9 / 10) \times 100 = 90.0\%$
- Loser percentage: $(1 / 10) \times 100 = 10.0\%$
- Verification: ✓ Correct

Deck Composition Verification

The program correctly implements the specified deck structure:

- Base deck: [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13]
- Each player deck: base_deck + base_deck = 26 cards
- Each value appears exactly twice per deck
- Both players receive identical deck compositions
- Fairness ensured through identical starting conditions

Part 5: Comprehensive Testing Strategy

Randomization Testing

The simulation's core feature is its random card shuffling mechanism, which requires thorough verification to ensure proper functionality. Multiple execution tests were conducted to confirm that the `random.shuffle()` function produces different card orders in each match. By running the simulation multiple times and recording the sequence of winners, we verified that outcomes vary appropriately across executions, demonstrating true randomization rather than predetermined patterns.

Sequential match independence was verified by examining whether the results of one match influenced subsequent matches. The deck shuffling occurs independently at the start of each match, ensuring that a player's victory or loss in one match has no mathematical impact on their probability of winning future matches. This independence is critical to maintaining game fairness.

The shuffle algorithm's effectiveness was evaluated through distribution analysis. Over multiple simulation runs, we observed that both players achieve approximately equal total win rates when averaged across many executions, confirming that neither player has a systematic advantage from the shuffling mechanism.

Logical Flow Verification

The round-by-round progression was traced through manual execution of several matches to verify correct card comparison logic. Each round correctly extracts cards from their respective deck positions (using zero-indexed array access with `round_num - 1`), compares the card values, and updates the appropriate win counter. This verification ensures that the array indexing never produces out-of-bounds errors or skips cards.

Match winner determination logic was tested through scenarios where players have different round win distributions. The program correctly identifies the player with more round wins as the match winner, properly handles exactly equal round wins as match ties, and accurately updates the match win counters after each match conclusion.

Part 5: Comprehensive Testing Strategy (Continued)

Statistical Accuracy Testing

Counter increment verification ensured that all statistical tracking variables update correctly throughout execution. Round win counters correctly increment from 0 to their final values (typically 10-16 per match), reset to 0 at the start of each new match, and accumulate properly within each match. Match win counters correctly increment when their respective player wins a match and maintain accurate totals across all 10 matches.

The tie counting mechanism was verified to correctly track both round ties (when cards are equal) and match ties (when round wins are equal). The `total_ties` variable accumulates all ties throughout the simulation, providing an accurate final count.

Win percentage calculation accuracy was verified through multiple test scenarios with known outcomes. The formula $(\text{match_wins} / 10) \times 100$ correctly produces percentages ranging from 0.0% to 100.0%, handles integer division appropriately by using floating-point arithmetic, and displays results with proper decimal precision.

Edge Case Testing

Perfect victory scenarios were tested by examining matches where one player wins all 26 rounds. The program correctly awards the match to the dominant player, updates statistics appropriately, and continues to the next match without errors.

Maximum tie scenarios were analyzed to ensure the program handles matches with unusually high numbers of tied rounds. Even when many rounds result in ties, the program correctly determines the match winner based on the remaining decisive rounds.

Zero-win scenarios were verified to ensure the program correctly handles situations where a player wins zero matches across all 10 games. The win percentage correctly calculates as 0.0%, and all statistical outputs remain accurate and properly formatted.

Part 6: Instructions for Use

Running the Simulation

The War Card Game simulation is designed for straightforward execution with no required user input during runtime. To run the simulation, navigate to the directory containing `war_card_game.py` and execute the command `python war_card_game.py` from the terminal or command prompt. The program will immediately begin the simulation and display real-time output as matches progress.

For users working in integrated development environments (IDEs) such as PyCharm, Visual Studio Code, or IDLE, the script can be opened in the editor and executed using the IDE's run functionality. The output will appear in the IDE's console window, providing the same detailed match-by-match information as command-line execution.

The simulation requires Python 3.x and uses only standard library modules (`random`), so no additional package installation is necessary. This ensures maximum compatibility across different Python environments and platforms.

Understanding the Output

The simulation produces structured output organized into three distinct phases: match initialization, round progression, and final statistics.

Match Initialization Phase:

Each match begins with a header displaying "`= Game X =`" where X is the match number (1-10). This header clearly delineates the boundaries between different matches and helps track simulation progress.

Round Progression Phase:

During each match, the program displays the outcome of every round. When a player wins a round, the output shows "Player X wins round count: Y" where Y is their current round win total within that match. This running count allows observers to track which player is leading during the match. When rounds result in ties, the output displays "Tie - no one wins this round" to clearly indicate that neither player gained an advantage.

Part 6: Instructions for Use (Continued)

Match Conclusion:

After all 26 rounds complete, the program announces the match winner with "Player X wins game!" or "Game is a tie!" if both players achieved equal round wins. This clear declaration makes it easy to track which player is winning more matches overall.

Final Statistics Phase:

Upon completing all 10 matches, the program displays comprehensive statistics:

- **Player 1 total wins:** Total number of matches won by Player 1 (0-10)
- **Player 2 total wins:** Total number of matches won by Player 2 (0-10)
- **Total ties:** Cumulative count of all ties (both round ties and match ties)
- **Player 1 win percentage:** Percentage of matches won by Player 1 (0.0-100.0%)
- **Player 2 win percentage:** Percentage of matches won by Player 2 (0.0-100.0%)

Interpreting Results

The final statistics provide insight into the overall balance of the simulation. In a truly random and fair game, win percentages should vary across multiple executions but average toward 50% for each player over many runs. Significant deviation from this pattern in a single run is normal due to the random nature of shuffling, but consistent bias across multiple executions would indicate a potential implementation issue.

The total ties count includes both individual round ties and complete match ties. Typically, this value ranges from 15-35 ties per simulation, though higher or lower values can occur due to random variation.

Sample Output Screenshots

Below are examples of the War Card Game simulation output showing the match progression and final statistics:

Game Output - Match Progression:

“/run/media/omniv1/T7/GCU_SWE_2023-2025/CST-180-Python/Photos/screenshot-2025-10-11_20-11-16.png” could not be found.

```
=== Game 9 ===
Player 1 wins round count: 1
Player 2 wins round count: 1
Player 1 wins round count: 2
Player 1 wins round count: 3
Player 1 wins round count: 4
Player 2 wins round count: 2
Player 1 wins round count: 5
Player 2 wins round count: 3
Player 1 wins round count: 6
Player 2 wins round count: 4
Player 2 wins round count: 5
Player 2 wins round count: 6
Player 1 wins round count: 7
Player 1 wins round count: 8
Player 2 wins round count: 7
Tie - no one wins this round
Player 1 wins round count: 9
Player 2 wins round count: 8
Tie - no one wins this round
Player 2 wins round count: 9
Player 1 wins round count: 10
Tie - no one wins this round
Player 1 wins round count: 11
Player 1 wins round count: 12
Player 2 wins round count: 10
Player 1 wins round count: 13
Player 1 wins game!

=== Game 10 ===
Player 1 wins round count: 1
Player 2 wins round count: 1
Player 2 wins round count: 2
Player 1 wins round count: 2
Player 1 wins round count: 3
Player 1 wins round count: 4
Player 2 wins round count: 3
Player 1 wins round count: 5
Player 2 wins round count: 4
Player 2 wins round count: 5
Player 1 wins round count: 6
Player 1 wins round count: 7
Player 1 wins round count: 8
Player 2 wins round count: 6
Player 1 wins round count: 9
Player 2 wins round count: 7
Player 1 wins round count: 10
Player 2 wins round count: 8
Player 2 wins round count: 9
Player 2 wins round count: 10
Player 2 wins round count: 11
Player 2 wins round count: 12
Player 2 wins round count: 13
Player 1 wins round count: 11
Player 1 wins round count: 12
Player 1 wins round count: 13
Game is a tie!
Player 1 total wins: 5
Player 2 total wins: 3
Total ties: 25
Player 1 win percentage: 50.0%
Player 2 win percentage: 30.0%
```

Game Output - Final Statistics:

“/run/media/omniv1/T7/GCU_SWE_2023-2025/CST-180-Python/Photos/image.png” could not be found.

```
=== Game 10 ===
Player 2 wins round count: 1
Tie - no one wins this round
Player 2 wins round count: 2
Player 1 wins round count: 1
Player 2 wins round count: 3
Player 2 wins round count: 4
Player 2 wins round count: 5
Player 1 wins round count: 2
Player 1 wins round count: 3
Player 1 wins round count: 4
Player 1 wins round count: 5
Player 2 wins round count: 6
Player 1 wins round count: 6
Player 2 wins round count: 7
Player 2 wins round count: 8
Player 1 wins round count: 7
Player 1 wins round count: 8
Player 1 wins round count: 9
Player 1 wins round count: 10
Player 1 wins round count: 11
Player 2 wins round count: 9
Player 1 wins round count: 12
Player 2 wins round count: 10
Player 1 wins round count: 13
Player 1 wins round count: 14
Player 1 wins round count: 15
Player 1 wins game!
Player 1 total wins: 6
Player 2 total wins: 2
Total ties: 20
Player 1 win percentage: 60.0%
Player 2 win percentage: 20.0%
```


These screenshots demonstrate the real-time output during game execution, showing how round wins accumulate, match winners are declared, and final statistics are displayed.

Modifying the Simulation

Advanced users can modify various aspects of the simulation by editing the source code:

- **Number of matches:** Change the range in `for match in range(1, 11):` (line 48) to alter the number of matches played
- **Deck composition:** Modify the `deck_of_cards` list (line 18) to change card values or distribution
- **Rounds per match:** Adjust the range in `for round_num in range(1, 27):` (line 66) to change the number of rounds per match (must match deck size)

Part 7: Code Quality and Best Practices

Documentation Standards

The War Card Game simulation demonstrates comprehensive documentation practices that facilitate code understanding and maintenance. Every significant code block includes pseudocode comments that explain both the algorithmic intent and the Python implementation, creating a parallel narrative that bridges theoretical design and practical execution.

Variable naming follows semantic conventions that reflect their purpose within the game context. Names like `player_1_deck`, `round_num`, `player_1_round_wins`, and `player_2_match_wins` immediately convey their meaning without requiring reference to comments or documentation. This self-documenting approach reduces cognitive load and makes the code accessible to programmers of varying experience levels.

The docstring at the file header provides high-level context about the program's purpose, functionality, and key rules. This overview allows readers to understand the program's goals before examining implementation details.

Inline comments serve multiple purposes: they explain complex logic, clarify the relationship between pseudocode and implementation, describe why certain operations are necessary (such as `round_num - 1` for zero-indexed array access), and mark the boundaries between different logical phases of the program.

Code Structure and Organization

The program employs a clear hierarchical structure with nested loops that mirror the game's conceptual organization. The outer loop represents the 10-match tournament structure, while the inner loop represents the 26 rounds within each match. This nesting pattern makes the program's flow intuitive and easy to trace.

Variable initialization occurs at appropriate scopes: match-level statistics (match wins) are initialized once at program start, while round-level statistics (round wins) are reset at the beginning of each match. This scoping ensures that counters accurately reflect their intended measurement periods.

Part 7: Code Quality and Best Practices (Continued)

Algorithm Efficiency

The simulation operates with optimal time complexity for its task. The nested loop structure results in $O(n \times m)$ complexity where n is the number of matches and m is the rounds per match ($10 \times 26 = 260$ operations). This linear scaling is appropriate for the problem domain and ensures near-instantaneous execution on modern hardware.

Memory usage remains minimal throughout execution. The program maintains only two 26-element arrays (the player decks) and several integer counters, resulting in $O(1)$ space complexity. No dynamic memory allocation or accumulation occurs during execution, preventing potential memory issues even with extended simulation runs.

Randomization Implementation

The program correctly uses Python's `random.shuffle()` function, which implements the Fisher-Yates shuffle algorithm. This approach provides cryptographically strong randomization suitable for game simulation purposes. The shuffle occurs independently for each player's deck at the start of each match, ensuring true randomness and preventing any correlation between match outcomes.

Testing and Validation Philosophy

The simulation's correctness can be verified through multiple approaches. Statistical validation over many executions should demonstrate approximately equal win rates for both players, confirming the absence of systematic bias. Single-execution correctness can be verified by confirming that match wins plus match ties equal 10 (the total number of matches), round wins plus round ties equal 26 for each match, and win percentages correctly calculate from match win totals.

Part 8: Conclusion

The War Card Game simulation successfully implements a complete game system that demonstrates core programming concepts while maintaining code clarity and correctness. The program accurately models the specified game rules, provides comprehensive real-time output, and calculates meaningful statistics about game outcomes.

The implementation follows best practices in code documentation, variable naming, and algorithmic structure. The extensive commenting and clear logical flow make the code accessible to students learning Python programming while demonstrating professional development standards.

The simulation's random nature ensures that each execution produces unique results, making it an engaging demonstration of probability and statistics in action. Over multiple runs, the program correctly demonstrates the expected statistical properties of a fair game, where both players have equal chances of victory.

Through comprehensive testing and validation, the program has been verified to correctly implement all specified requirements, handle edge cases appropriately, and produce mathematically accurate results. The clear output format makes it easy to understand game progression and final outcomes.